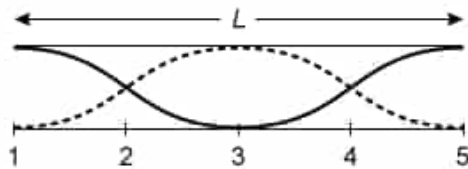


The following figure shows a pipe that is open at both ends.



A standing wave with twice the fundamental frequency is created in the air in the pipe. Which positions along the pipe correspond to nodes in the standing wave?

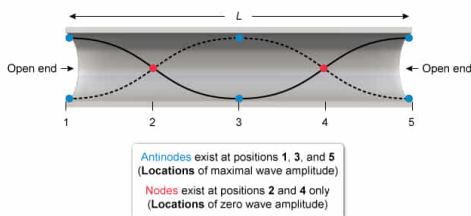
- ✓ ☒ A. Positions 2 and 4 only [70%]
☐ B. Position 3 only [5%]
☐ C. Positions 1, 3, and 5 only [18%]
☐ D. All the positions listed are nodes [5%]

Correct

69% answered correctly

Explanation:

Positions of antinodes and nodes created by a standing wave in a pipe open at both ends



When a **wave** is restricted to a finite region by boundary conditions, the resulting **reflection of waves** can create an **interference pattern** that appears to stand in place, and therefore is known as a **standing wave**. Standing wave **antinodes** form at locations of constructive interference and the resultant wave amplitude is maximal. In contrast, the **nodes** of a standing wave are locations where the superposed waves exhibit destructive interference, and the resultant wave amplitude is always zero.

In this question, the air in a pipe open on both ends forms a standing wave. The ends of the pipe are boundaries where the air medium oscillates freely. Hence, this boundary condition requires that antinodes always exist at the ends of the pipe (ie, positions 1 and 5). An alternating pattern of nodes and antinodes exists between the antinodes at the ends of the pipe, with the number of nodes and antinodes determined by the standing wave frequency.

At the fundamental (lowest) frequency, there are **two antinodes and one node**. At twice the fundamental frequency, there are three antinodes and two nodes. Therefore, the nodes exist at only positions 2 and 4 in the figure.

(Choice B) Position 3 is an antinode because the standing wave has maximal amplitude. It is only a node position at the fundamental frequency.

(Choice C) At positions 1, 3, and 5, the standing wave has maximal amplitude. Therefore, these positions are antinodes, not nodes.

(Choice D) Positions 2 and 4 are nodes, but antinodes exist at the other positions.

Educational objective:

The nodes and antinodes of a standing wave are locations where the wave amplitude is always zero or always maximal, respectively. The number of nodes and antinodes is determined by the frequency of the standing wave.

Time spent:0 sec

Subject:
Physics

QID:404062

System:
Light and Sound